

The Subtle Yet Impactful Choices in Matching-Adjusted Indirect Comparison for TTE endpoints: Insights from Simulation

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Introduction

Matching-Adjusted Indirect Comparison (MAIC) reweights individual patient data (IPD) to match aggregate data (AGD) from comparator studies, enabling indirect comparisons when head-to-head trials are unavailable. While conceptually straightforward, MAIC implementation involves numerous subtle choices—weight normalization schemes, estimation methods, and variance estimators—that can have impact on results.

Objectives

- Systematically compare twelve weighting approaches combining MAIC, maximum effective sample size (MEW), and inverse probability of treatment weighting (IPTW) with four normalization schemes. IPTW serves as a methodological benchmark—the "best case" if IPD were available from both studies—quantifying the efficiency cost of having only aggregate external data.
- Evaluate finite-sample properties of four confidence interval construction methods
- Properly account for non-collapsibility by computing true marginal hazard ratios
- Provide evidence-based recommendations for method selection in practice

Methods

Target Estimand and Non-collapsibility

The target estimand is typically the Average Treatment Effect in the Untreated (ATUT):

$$\theta_{\text{ATUT}} = E[Y(1) - Y(0)|T = 0] \quad (1)$$

where $T = 0$ indicates the external (control) population. ATUT is the natural target for MAIC applications because the external trial typically contains the comparator treatment that serves as the reference for decision-making. We have IPD from the internal trial (sponsor's investigational treatment) and only aggregate data from the external trial (established comparator), making ATUT the estimand of interest for HTA submissions.

For time-to-event outcomes with Cox regression, the marginal HR differs from conditional HR due to non-collapsibility.

Weight Estimation Methods

All methods reweight the internal (treated) population to match the external (control) population distribution. Let $T_i \in \{0, 1\}$ denote study membership (0=external, 1=internal), $\mathbf{X}_i$ be baseline covariates, and $\bar{\mathbf{X}}_{\text{external}}$ be the mean covariate vector in the external population. Weights w_i are applied only to internal subjects; external subjects receive weight = 1 as the target population.

- MAIC weights:** Method of moments optimization to balance covariate means:

$$\min_{\alpha} \sum_{i:T=1} \exp(\alpha^T \mathbf{X}_i^c), \quad \text{where } \mathbf{X}_i^c = \mathbf{X}_i - \bar{\mathbf{X}}_{\text{external}}$$

Results in weights $w_i = \exp(\hat{\alpha}^T \mathbf{X}_i^c)$ ensuring exact balance: $\sum w_i \mathbf{X}_i / \sum w_i = \bar{\mathbf{X}}_{\text{external}}$

- MEW weights:** Maximize effective sample size (ESS) while achieving exact covariate balance:

$$\max_w \text{ESS}(\mathbf{w}) = \frac{(\sum w_i)^2}{\sum w_i^2}, \quad \text{s.t. } \sum w_i \mathbf{X}_i = n_1 \bar{\mathbf{X}}_{\text{external}}$$

Closed-form solution: $\hat{\mathbf{w}} = \mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{b}$, where $\mathbf{A} = [\mathbf{1}, \mathbf{X}]$ and $\mathbf{b} = [n_1, n_1 \bar{\mathbf{X}}_{\text{external}}^T]^T$

- IPTW ATUT weights:** Model-based propensity score weights targeting ATUT:

$$w_i = \frac{P(T = 0 | \mathbf{X}_i)}{P(T = 1 | \mathbf{X}_i)} = \frac{1 - e(\mathbf{X}_i)}{e(\mathbf{X}_i)}$$

where $e(\mathbf{X}) = P(T = 1 | \mathbf{X})$ is the propensity score estimated via logistic regression

Normalization Schemes

Four schemes evaluated: Original (OW), Sum-to-one (SW1), Sum-to-N (SWN), Sum-to-ESS (SWESS)

Simulation Design

Data Generation Mechanism

- Baseline covariates:**
 - External population: $X_1 \sim \mathcal{N}(0.5, 0.5^2)$, $X_j \sim \text{Bernoulli}(0.5)$ for $j = 2, 3, 4$
 - Internal population: Means scaled by parameter d to control population differences
 - $X_1 \sim \mathcal{N}(0.5d, 0.5^2)$, $X_j \sim \text{Bernoulli}(\min(0.5d, 1))$
 - Parameter d calibrated via Monte Carlo to achieve target ESS ratios
- Survival times:** Proportional hazards model with Weibull or log-normal baseline hazard:

$$h(t | \mathbf{X}, Z) = h_0(t) \exp(\beta^T \mathbf{X} + \gamma Z) \quad (2)$$

Baseline hazard parameters calibrated to achieve specified median survival (24 months) and 48-month survival (0.15)

- Censoring:** Realistic dropout pattern with 60% early (uniform 0-6 months) and 40% late (uniform 6-48 months), plus administrative censoring at 48 months

Computing True Marginal ATUT HR

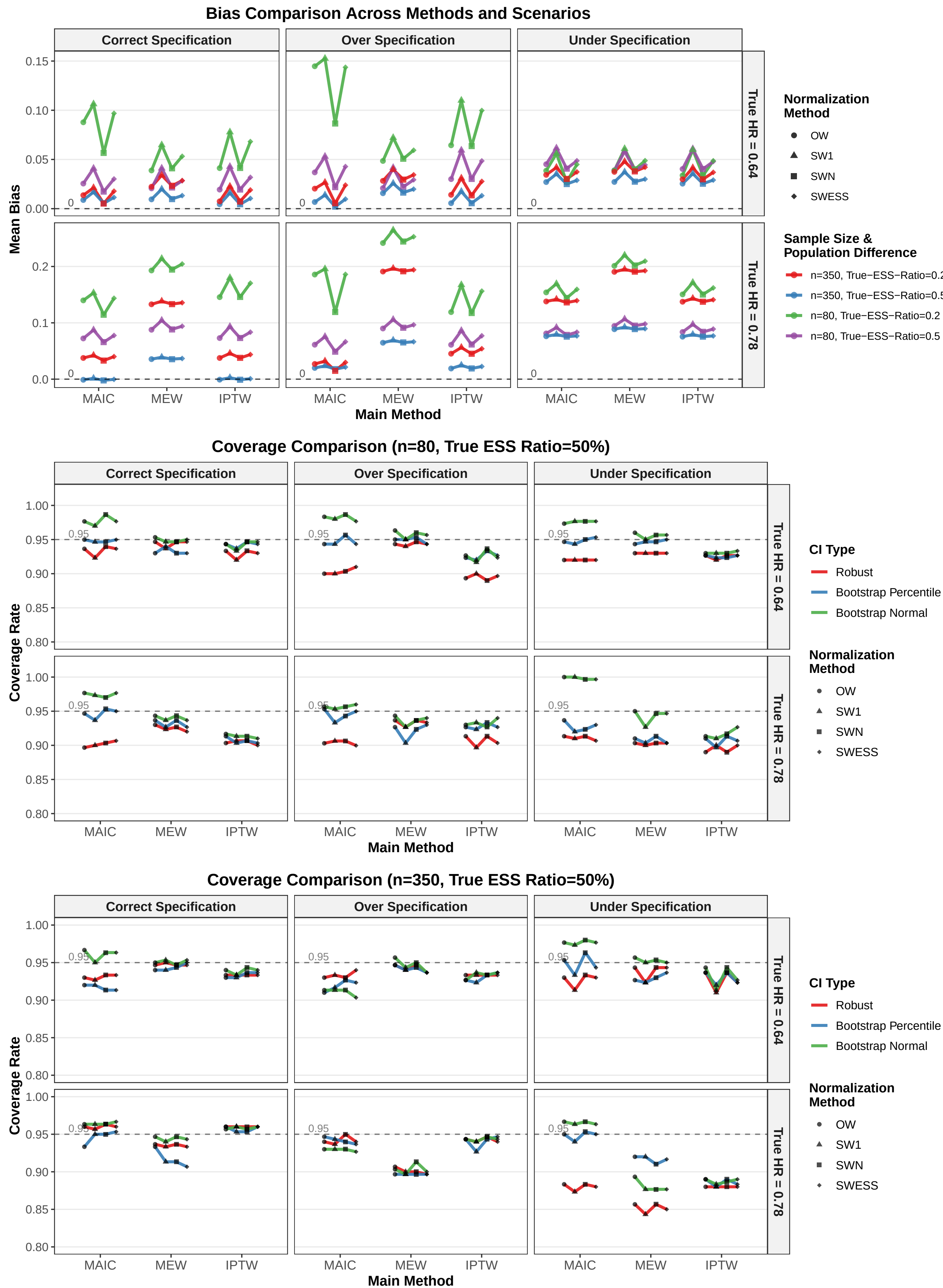
The true marginal ATUT HR can not be directly set, but can be evaluated via Monte Carlo simulation with 10,000 samples. This procedure generates a large sample from the external population, randomly assigns treatments, applies realistic censoring patterns, and fits an unweighted Cox model to obtain the marginal HR. This approach properly accounts for non-collapsibility, censoring mechanisms, and finite follow-up, providing the correct benchmark against which to evaluate bias—avoiding the fundamental error of comparing estimates to the wrong (conditional) target parameter.

Simulation Scenarios and Results

Table 1: Factorial Simulation Design

Parameter	Values	Parameter	Values
Sample size (per arm)	80, 350	Dropout rate	0.20
Conditional HR	0.75	Median survival	24 months
Distribution	Weibull, Log-normal	48-month survival	0.15
Event rate (control)	0.70	Max follow-up	48 months
ESS ratio	0.20, 0.50	Replications	300
Model specification	Correct, Under, Over	Bootstrap samples	500

Total scenarios: $24 (2 \times 1 \times 2 \times 1 \times 2 \times 3)$ factorial design with model specification: **Correct:** All prognostic factors (X1-X3); **Under:** Omit X3; **Over:** Include non-prognostic X4



Conclusions

Even with this example subset of scenarios, we observe that method selection can impact both bias and coverage probability, with some approaches performing optimally only under specific sample size and overlap conditions. These findings highlight the critical importance of understanding and transparently reporting all methodological choices in MAIC applications. A comprehensive examination covering a wider variety of scenarios will be available in our forthcoming manuscript—please contact the author team for further discussion and more preliminary results.

